

Task 2 – Basic Firewall Configuration with UFW

Oasis Infobyte Cyber Security Internship

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Track: Cyber Security

Task: Task 2 – Basic Firewall Configuration with UFW

1. Introduction

This task was completed as part of my Oasis Infobyte Cyber Security Internship. The objective was to configure a basic firewall using UFW (Uncomplicated Firewall) on Kali Linux. I created rules to control incoming network traffic, enabled the firewall, and verified the final configuration.

2. Objective

The main objective was to understand the basic use of a Linux firewall. I configured default policies, allowed SSH through port 22, denied HTTP through port 80, added HTTPS and DNS rules, enabled UFW, and verified the configuration.

3. Environment Used

Operating System: Kali Linux

Firewall: UFW (Uncomplicated Firewall)

Shell: Bash

Testing tool: curl

4. UFW Installation

UFW was installed on the Kali Linux system using the package manager. After installation, I checked the firewall status. Initially, the firewall was inactive because it had not yet been enabled.

5. Default Firewall Policies

I configured the default policies so that incoming connections were denied while outgoing connections were allowed.

```
ufw default deny incoming
ufw default allow outgoing
```

6. Firewall Rules Configured

SSH – Port 22: allowed using `ufw allow 22/tcp`.

HTTP – Port 80: denied using `ufw deny 80/tcp`.

HTTPS – Port 443: allowed using `ufw allow 443/tcp`.

DNS – Port 53: allowed using `ufw allow 53`.

7. Enabling the Firewall

After configuring the rules, UFW was enabled and became active on the Kali Linux system.

8. Verification

The final configuration was verified using `ufw status verbose`. The output showed the default policies and the configured SSH, HTTP, HTTPS, and DNS rules.

9. HTTP Traffic Test

I performed a local HTTP test using `curl -I http://127.0.0.1`. The UFW configuration was also checked to confirm that TCP port 80 had a deny rule. A connection failure alone was not treated as proof that UFW caused the failure because a local web service may not have been running.

10. Security Analysis

The configuration provides basic control over network exposure. The default deny incoming policy blocks unsolicited incoming connections unless they are explicitly allowed. SSH was allowed for administration, HTTP was denied for this task, and HTTPS and DNS were added as additional rules.

11. Recommendations

Allow only required ports and services. Keep unnecessary services disabled. Review firewall rules regularly. Prefer secure protocols such as HTTPS where appropriate. Keep the operating system and network services updated.

12. Ethical Considerations

All configuration and testing was performed on my own Kali Linux environment for educational purposes. No unauthorized external systems or networks were tested.

13. Conclusion

This task gave me practical experience with UFW and basic Linux firewall configuration. I learned how to set default policies, create allow and deny rules, enable the firewall, verify the configuration, and perform a basic traffic test.

14. Evidence / Screenshots

The final GitHub documentation contains the actual Task 2 screenshots covering the UFW configuration/status and the UFW script plus HTTP test.